

Ethan Gates

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SUMMARY

AI-Native Software Engineer building production AI products, agent systems, and data-intensive full-stack applications. Healthcare SaaS engineer and M.S. Computer Science (AI/ML) student with hands-on work in grounded AI, local inference, orchestration, and evaluation.

EDUCATION

M.S. Computer Science , Western Governors University <i>Artificial Intelligence and Machine Learning Specialization</i>	Expected: Feb 2027
B.S. Computer Science , Western Governors University	Completed: Feb 2025

PROFESSIONAL EXPERIENCE

Software Engineer , DOVAXIS (Healthcare SaaS patient referrals & admissions) Began as a paid engineering intern; converted to full-time November 2025.	May 2025 – Present
- Built governed document-intelligence workflows with lifecycle/version controls, curated regulation search, grounded AI answers with citations, retrieval debugging, and usage/cost analytics. - Consolidated scattered role and resource checks into a Permit.io-backed authorization facade with bulk decisions and cached permission loading across routes, navigation, folders, documents, eForms, and regulations. - Designed, executed, and audited a staged cutover that moved every customer organization onto a new authorization model in two days with no customer-facing downtime. - Co-owned end-to-end test automation: authored the suite's governance layer (test-metadata contract, standards doc, five static gate scripts, generated inventory) and led a cull-and-rebuild that cut the Playwright suite from 68 specs to 36, then rebuilt it to 62 governed specs. - Authored 256 PRs with 225 merged in the first year, spanning AI features, authorization, clinical workflows, tenant rollout, and quality infrastructure.	

INDEPENDENT PROJECTS

- GatesAI Chat**: Built and shipped a local-first AI workspace (React/Tauri/Rust with a Go bridge) featuring OpenRouter/Ollama routing, multi-round tool execution, sandboxed files/shell/Git/SQLite, memory, MCP, and image generation, backed by 995 unit/component and 20 end-to-end tests. [\[Repo\]](#)
- Agent Orchestrator**: Architected a TypeScript multi-agent system that dispatches coding agents across isolated Git worktree lanes with path ownership, deterministic build/test gates, bounded campaigns, integration workflows, event history, and model/strategy benchmarking. Private repo; public release of the proven subset in progress.
- Fathom**: Built a market-analysis suite with 155 years of asset-class history, portfolio backtesting, Monte Carlo retirement simulation, SEC-filing research, and broker-CSV portfolio X-ray as a pure-TypeScript analytics engine with a React frontend. [\[Demo\]](#)
- N2K Almanac**: Engineered a TypeScript solver computing the optimal equation for 370K+ dice/target combinations, encoded in a custom compact binary format and served through a Web Worker for instant lookups, with a React/MobX data-visualization frontend. [\[Demo\]](#)

TECHNICAL SKILLS

- Languages**: TypeScript, Python, Rust, JavaScript, SQL, C#.
- AI / LLM Engineering**: Agent orchestration, tool calling, grounded retrieval/citations, embeddings, evaluation & benchmarking, OpenRouter, Ollama, MCP.
- Frameworks & Libraries**: React, MobX, FastAPI, Tauri, TensorFlow, NumPy, Pandas, D3.js, Tailwind CSS.
- Tools & Infrastructure**: Git/GitHub, Playwright, Firebase, PostgreSQL, MySQL, Docker, Vite, Google Cloud Run, Linux.